

DAY 1 WEEK 1

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GROUP:S2

Day 1: Problem Understanding & System Definition

1. Problem Restatement

Electric vehicles are becoming mainstream, but "range anxiety" remains a significant barrier for users. This project addresses this by building a machine learning model that accurately predicts two critical battery metrics:

- State of Charge (SoC): The battery's current fuel gauge (0-100%).
- Remaining Charging Time: The time in minutes until the battery reaches full charge.

Instead of relying on complex physics-based battery models, I will use historical sensor data from charging sessions. By training ML models on patterns in voltage, current, power, and temperature, I aim to create a reliable software-only solution that could eventually help improve user experience and battery management systems.

2. End-to-End Software Pipeline

The software-only pipeline for this project is:

1. Data Loading: Load the provided CSV files (SOC DATA SET.zip).
2. Data Quality Assessment: Check for missing values, outliers, and inconsistencies.
3. Data Cleaning & Preprocessing: Handle missing data, remove outliers, and ensure data integrity.
4. Exploratory Data Analysis (EDA): Visualize distributions, correlations, and relationships between features and targets.

5. Feature Engineering: Create domain-relevant features from raw data (e.g., voltage slope, rolling averages).
6. Modeling: Train and compare Linear Regression (baseline), Random Forest, XGBoost, ANN.
7. Evaluation: Assess model performance using MAE, RMSE, and R^2 for both targets.
8. Analysis: Perform feature importance analysis and error analysis across charging phases.

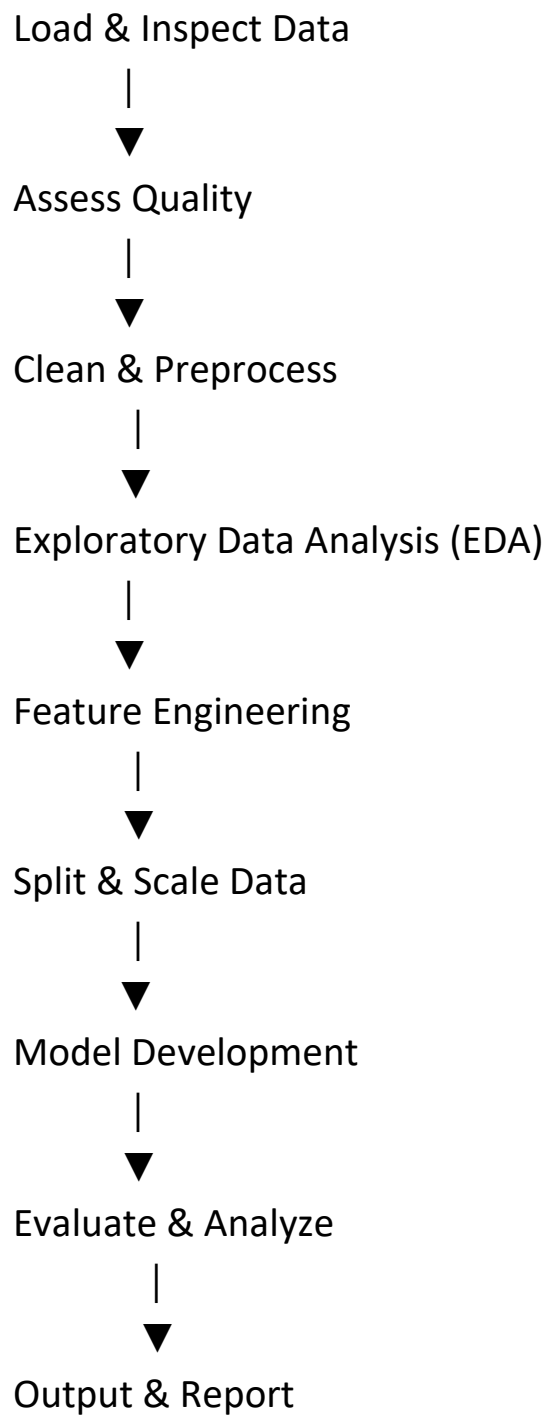
3. Success Criteria

- Develop ML models that predict SoC and Remaining Charging Time with reasonable accuracy.
- Provide a clear comparison between baseline and advanced models.
- Identify the most influential features for each prediction target.
- Create a complete, well-documented, and reproducible analysis in Python.

4. Key Assumptions

1. The provided dataset is a representative sample of real-world EV charging behavior.
2. The data is mostly accurate without systematic sensor errors.
3. We are modeling a single, typical battery type (likely Lithium-ion).
4. The "Remaining Charging Time" target is time to full charge from the current state.

Flow Chart:



5. Questions for Mentor

1. What is the origin of the dataset (lab tests, real-world vehicles)?
2. What is the sampling frequency of the measurements (1 sec, 1 min, etc.)?
3. Is there any data on battery age/health/cycle number?
4. How should we handle charging sessions that don't reach 100% SoC?
5. Is the charging power constant or variable (CC-CV profile)?
6. What is the battery chemistry (e.g., Li-ion, LiFePO₄)?